

Proof of work (PoW). Originally advocated by Back in 2002, the consensus mechanism for the two leading *blockchains*: *Bitcoin* and *Ethereum*. *Miners* compete to find a rare *cryptographic hash*, which is hard to find but easy to verify. Miners are rewarded for finding the cryptographic hash and using it to add a block to the *blockchain*. The computing difficulty of finding the hash makes it impractical to go backward to rewrite the history of a leading blockchain.

Router contracts. In the context of *decentralized exchange*, a contract that determines the most efficient path of swaps to get the lowest slippage, if no direct trading pair is available on, for example, Uniswap.

Scaling risk. The limited ability of most current blockchains to handle a larger number of transactions per second. See *vertical scaling* and *horizontal scaling*.

Schelling-point oracle. A type of *oracle* that relies on the owners of a fixed supply of tokens to vote on the outcome of an event or report a price of an asset.

Sharding. A process of horizontally splitting a database, in the context of a blockchain. Also known as *horizontal scaling*. Divides the work of the system into multiple pieces, retaining decentralization, but increasing the throughput of the system through parallelization. *Ethereum 2.0* takes this approach with the goal of reducing network congestion and increasing the number of transactions per second.

Slashing. A mechanism in *proof of stake blockchain* protocols intended to discourage certain user misbehavior.